Theory of an Amplified Closed-Sagnac-Loop Interferometric Fiber-Optic Gyroscope

Chin-Yi Liaw, Yan Zhou, and Yee-Loy Lam, *Member, IEEE*

***Abstract*— An amplified closed-Sagnac loop interferometric fiber optic gyroscope (AC-IFOG) has been proposed and demonstrated in our earlier work. This paper reports the theoretical analysis of its operation. The analysis has been focused on the effect of the erbium-doped fiber amplifier (EDFA) inserted within the reentrant path. The analysis shows that the sensitivity can be improved by at least two orders of magnitude as compared to a conventional IFOG if the EDFA is an ideal device. The simulated open-loop response coincides well with that of the experimental data. Further analysis shows that the detection limit is restricted by the beat noise that build ups rapidly in successive circulations. The rate of noise build-up depends greatly on the optical bandwidth of the closed-Sagnac loop.**

***Index Terms*—Active reentrant fiber gyroscope, erbium-doped fiber amplifier, fiber-optic sensors, Sagnac effect.**

I. Introduction

CONVENTIONAL interferometric fiber-optic rotation sensors propagate light once around a multiturn optical fiber coil [1]. As the phase modulation does not provide amplification, the device operates passively. System noise and the sensing coil length limit the rotation detection sensitivity [2]. In the case where high-precision detection capability is required, a long fiber coil with a length of a few kilometers is essential for boosting the Sagnac effect. To circumvent the long fiber requirement, active interferometric fiber-optic gyroscopes (IFOG's) have been experimented with [3], [4]. Generally, they make use of the active recirculating delay line (ARDL) to recirculate the light to accumulate the Sagnac-induced phase shift more than once within the sensing loop. An optical amplifier is generally incorporated within the loop to compensate for the loss of optical power. Two typical configurations of IFOG's using optical amplifiers are the amplified fiber ring resonator gyroscope [5] and the amplified reentrant fiber gyroscope [3].

A. Proposal of the Amplified Closed-Sagnac-Loop Interferometric Fiber-Optic Gyroscope

Although the amplified reentrant fiber gyroscope has a built-in linear scale factor with frequency readout, it requires a geometrically long fiber (>10 km) to detect a low rotation rate (0.046 rad/s or 9488°/hr). The storage time is limited by the mechanical chopper and the rapid amplified spontaneous emission (ASE) accumulated during each recirculation. In order to overcome these problems, we have designed and investigated a novel reentrant interferometric fiber-optic gyroscope [6]. Although it is based on a working principle similar to that of the reentrant fiber-optic gyroscopes, it is, however, implemented in a completely different manner. With two additional couplers, a bidirectional erbium-doped fiber amplifier (EDFA) is inserted in the fiber loop to bypass the two arms of the splitter/mixer coupler (coupler 2 of Fig. 1), and it links the two ends of the fiber coil together. As a result, the multiturn optical fiber coil is closed by the insertion of this EDFA. This configuration is totally different from the previously reported active reentrant fiber gyroscope design [7], [8], which uses only one coupler and two Sagnac loops instead of one. Also, as a departure from previous active reentrant fiber gyroscope operation, which is based on pulse propagation, both pulse mode and CW operations are possible for the new design and are thus investigated simultaneously . In this paper, our new configuration is referred to as amplified closed-Sagnac-loop interferometric fiber optic gyroscope (AC-IFOG).

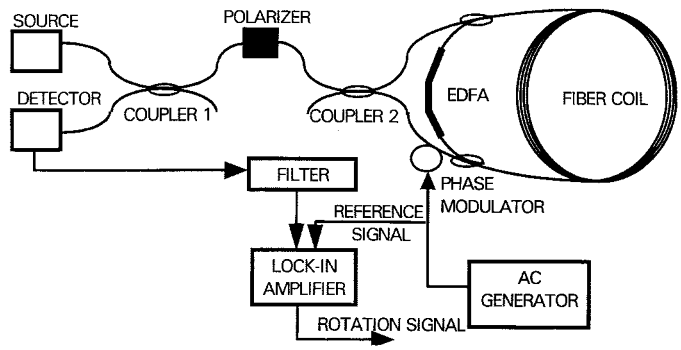


Fig. 1. Schematic of the proposed AC-IFOG.

For the AC-IFOG, we assume a simple case where two EDFA couplers have exactly the same coupling ratio. Redirection of light occurs when part of the counterpropagating waves are coupled into the erbium-doped fiber before they return to the splitter/mixer coupler. These two redirected light waves will be amplified and subsequently return to the sensing coil via the other EDFA coupler. The recirculation of light through the multiturn optical fiber coil causes the Sagnac phase shift induced in each and every circulation to be accumulated. The total Sagnac phase shift between a pair of counterpropagating waves is therefore proportional to the number of recirculation m. If a low coherence light source is used, then only the wave pairs that have recirculated the same number of times in the

Manuscript received March 31, 1999.
The authors are with the School of Electrical and Electronic Engineering, Nanyang Technological University, 639798 Singapore.
Publisher Item Identifier S 0018-9197(99)09431-2.

loop will interfere with each other. In this case, the rotation rate detection scheme has to be interference based in order to give an output attributable to the nonreciprocal phase shift of the counter-propagating waves. The amplification provided by the EDFA should ideally overcome the loss in the light intensity during each recirculation.

B. Mode of Operation

There are two ways of operating the proposed gyroscope, depending on whether the laser source is modulated (pulse mode) or not (CW mode).

1) Operating in Pulse Mode: When in pulse mode, the operation can be explained by considering a single light pulse with its duration shorter than the loop transit time (T_{loop}). The pulse splits into two counterpropagating pulses inside the Sagnac interferometer. Thereafter, some of these signals are coupled into the EDFA, where they recirculate again in the multiturn sensing coil for the duration (T_{loop}). After each recirculation, a fraction of each pulse exits the closed sensing loop and recombines with the other coherent partner through the Sagnac-loop splitter/mixer coupler. It is noted that every single pulse from the laser source will give rise to a train of returning pulses at the detector, with the pulse spacing equal to the loop transit time T_{loop}. In the absence of rotation ($\Omega = 0$), Sagnac interferometer reciprocity tells us that the pulse train envelope is constant and has maximum amplitude (assuming that the gain of the EDFA just compensates for the overall loss of each recirculation). Rotation ($\Omega \neq 0$) creates a nonreciprocal phase shift $\delta\phi$ between the first two counterpropagating wave pairs. As the number of circulation m increases, the cumulative phase shift $m\delta\phi$ also increases with the effective loop length, and the envelope of the interfered intensity is sinusoidally modulated, with a frequency f proportional to the rotation rate Ω.

The rotation rate can be determined from the frequency of the envelope as [7]

$$f = \frac{D}{n_f \lambda} \cdot \Omega \tag{1}$$

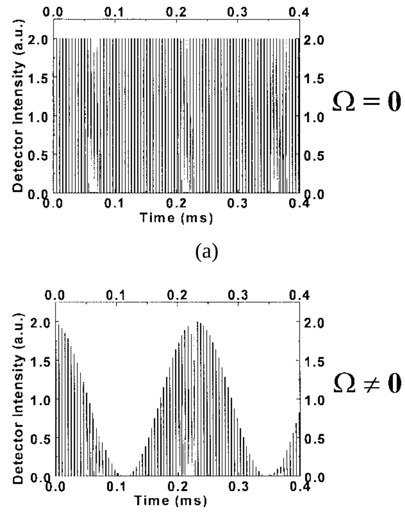
where λ is the light wavelength in free space, n_f is the refractive index of the fiber core, D is the loop diameter, and Ω is the rotation rate. An electronic spectrum analyzer can be used to measure the frequency f.

Mathematically, after m round trips of signal recirculation, the output signal envelope is proportional to

$$A_{\text{Envelope}} = 1 + \cos[2\pi f m T_{\text{loop}}]. \tag{2}$$

This is deduced from

$$\begin{aligned} I_{\text{out}} &\sim 1 + \cos(m\Delta\phi) \\ &= 1 + \cos\left[2\pi m \frac{LD}{\lambda c}\,\Omega\right], \quad \text{since } \Delta\phi = \frac{2\pi LD}{\lambda c}\,\Omega \\ &= 1 + \cos\left[2\pi m \frac{n_f LD}{n_f \lambda c}\,\Omega\right] \\ &= 1 + \cos\left[2\pi m \frac{D\Omega}{n_f\lambda} \times \frac{n_f L}{c}\right] \\ &= 1 + \cos\left[2\pi m f T_{\text{loop}}\right] \end{aligned}$$



(a)

(b)

Fig. 2. Reciprocal output when $G = L$. (a) Without rotation. (b) With rotation.

where

$$f = \frac{D\Omega}{n_f \lambda} \text{ and } T_{\text{loop}} = \frac{n_f L}{c}$$

where f is the envelope modulation frequency, m is the number of signal recirculation, and T_{loop} is the loop transit time.

Several computer simulations have been carried out based on (2), for different Ω, using $n_f = 1.46$, $\lambda = 1.55$ μm, $D = 10$ cm, $L = 1000$ m, $c = 3 \times 10^8$ m/s and a pulsewidth of 1 μs. The simulation results with $\Omega = 0$ and $\Omega = 0.1$ rad/s are shown in Fig. 2(a) and (b), respectively. These are for the case where the recirculation loss L is just compensated for by the amplifier gain G, i.e., when $G = L$. If $G < L$, the response dies off rapidly at an exponential rate, and this, in fact, has been experimentally demonstrated [7]. Despite the difference in the envelope of the signal intensity, the frequency of the sinusoidal component is still proportional to the rotation rate as given in (1).

2) Operating in CW Mode with EDFA Switching: It was reported in previous studies [8] that, to achieve a sensitivity comparable to that for an active reentrant fiber gyroscope or AC-IFOG to that of the best IFOG, a configuration with a relatively large dimension ($D = 10$ km) or a large optical delay (in milliseconds) is necessary. For this reason, the AC-IFOG operating in pulse mode is considered to be impractical and expensive. An alternative approach is to operate the AC-IFOG in CW mode instead of pulse mode. In the CW mode, the intensity of the laser source is not modulated. Considering the Sagnac effect alone, we expect that increasing the number of signal circulation can indefinitely increase the AC-IFOG sensitivity. But this cannot be achieved in a real device because of ASE noise that is built up during recirculation and the pump intensity noise. Therefore, a feasible operating technique is to have the EDFA turned on for a specific duration to restrict the number of recirculations. Assuming that this can be carried out, the analytical expression for the output of the AC-IFOG can be derived. While detailed derivation can be referred to in our previous paper [6], a simplified equation, representing the output of the AC-IFOG in CW mode right after the nth recirculation, can be given by

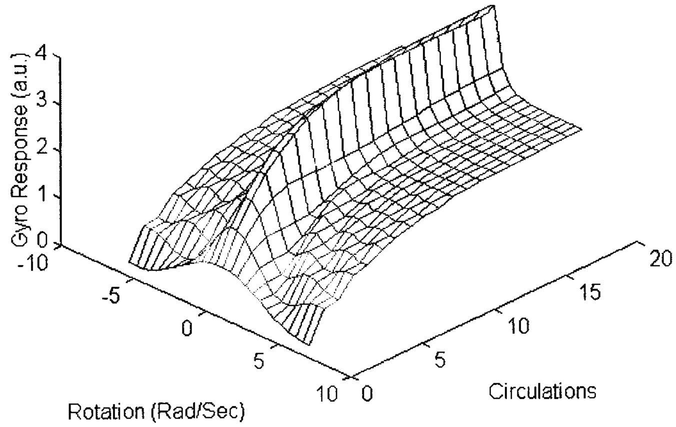
$$T(\Delta\phi_R) = \tfrac{1}{2}\, t_o^2 \sum_{m=1}^{n} A_{RC}^{2m}[1+\cos(m\Delta\phi_R)] \tag{3}$$

where $T(\Delta\phi_R)$ represents the output/input transmission ratio of light exiting/entering the Sagnac loop. The net amount of amplification (or loss) realized for each round trip (A_{RC}) is equal to the gain of the EDFA less its insertion loss (due to the WDM's and the couplers) and losses in the sensing coil. The variable m represents the recirculation number and n represents the ON-duration of the EDFA measured in the number of trips around the loop. The parameter t_o^2 is a transmission coefficient (or constant) and $\Delta\phi_R$ is the single round-trip Sagnac phase shift.

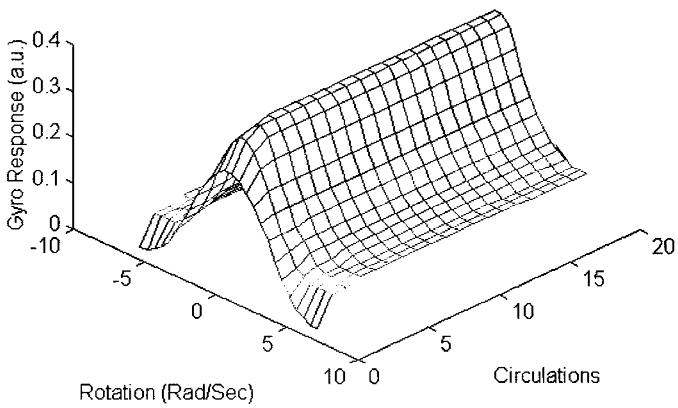
A series of simulations have been performed on this new configuration using (3). For the simulation, we have used a coil length of 200 m with a loop diameter of 10 cm, making the loop transit or circulation time ($\tau =$) 1 μs. In Fig. 3(a), the response of the gyroscope is plotted as a function of the number of recirculations, when the net single round-trip gain A_{RC} is 0.9. It can be seen that, as the number of recirculation increases, the overall response curve becomes narrower/sharper and its peak value also increases. In fact, the simulation shows that the peak can reach as high as eight times that of conventional IFOG's. Based on (3), it can be found that, when A_{RC} is 0.1, the response does not change much even if the number of recirculations is significantly increased, as can be seen in Fig. 3(b). In this case, the response curves have low peaks and are broad. The simulation shows that the single round-trip gain A_{RC} controls mainly the maximum achievable finesse of the gyroscope response. The greater the A_{RC}, the faster it is to achieve a higher finesse response.

It should be pointed out that, in the above simulation, it is assumed that the optical amplifier also acts as an ideal optical gate such that the path of the EDFA is broken when the EDFA is off.

3) Sensitivity of CW EDFA Operation: The sensitivity in this case refers to the minimum detectable relative change in the transmission ratio of the gyro with respect to a corresponding change in the rotation-induced phase shift. This is usually highest at the $\pi/2$ point on the response curve for



(a)



(b)

Fig. 3. Gyroscope response with different EDFA's for durations (a) $A_{RC} = 0.9$ and (b) $A_{RC} = 0.1$.

a conventional IFOG. Using this terminology, the sensitivity of a gyroscope is defined as $S(\Delta\phi_R) = dT(\Delta\phi_R)/d\Delta\phi_R$ [9]. To determine the sensitivity of the proposed AC-IFOG, the gyro output $T(\Delta\phi_R)$ in (3) can be expressed in (4), shown at the bottom of the page, using the relation $\sum_{m=1}^{n} A^m = (A - A^{n+1}/1 - A)$. By differentiation, we obtain the equation for sensitivity as (5), shown at the bottom of the page, where $a = 2A_{RC}^2/(1+A_{RC}^4)$. From (5), the optimum $\Delta\phi_R$ that makes $S(\Delta\phi_R)$ maximum is found by differentiating and setting $dS(\Delta\phi_{\rm opt})/d\Delta\phi_{\rm opt} = 0$. The resultant equation is lengthy, but it can be solved numerically

$$\begin{aligned} T(\Delta\phi_R) = \frac{1}{2}t_o^2\Bigg\{ &\frac{A_{RC}^2 - A_{RC}^{2(n+1)}}{1-A_{RC}^2} + \frac{A_{RC}^2\cos(\Delta\phi_R) - A_{RC}^4}{1-2A_{RC}^2\cos(\Delta\phi_R)+A_{RC}^4} \\ &+ \frac{A_{RC}^{2(n+2)}\cos(n\Delta\phi_R) - A_{RC}^{2(n+1)}\cos[(n+1)\Delta\phi_R]}{1-2A_{RC}^2\cos(\Delta\phi_R)+A_{RC}^4}\Bigg\} \end{aligned} \tag{4}$$

$$\begin{aligned} S(\Delta\phi_R) = \frac{1}{2}t_o^2\Bigg\{ &\frac{(A_{RC}^4-1)}{(1+A_{RC}^4)}\frac{a\sin(\Delta\phi_R)}{[1-a\cos(\Delta\phi_R)]^2} + \frac{an}{2}A_{RC}^{2(n+1)}\frac{[a\sin((n-1)\Delta\phi_R) - \sin(n\Delta\phi_R)]}{[1-a\cos(\Delta\phi_R)]^2} \\ &- \frac{a}{2}A_{RC}^{2n}\frac{[a\sin(n\Delta\phi_R) + (an\cos(n\Delta\phi_R) - (n+1))\sin((n+1)\Delta\phi_R)]}{[1-a\cos(\Delta\phi_R)]^2}\Bigg\} \end{aligned} \tag{5}$$

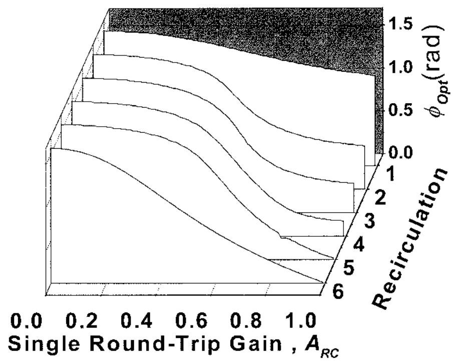


Fig. 4. Optimum phase biases versus single round-trip gain A_{RC}.

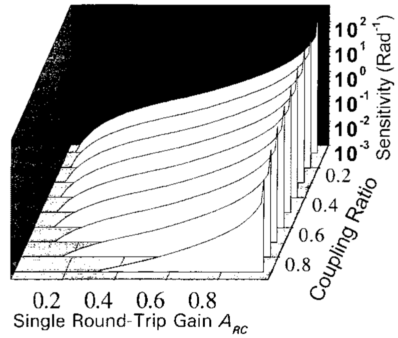


Fig. 5. Maximum sensitivity as a function of round-trip gain and coupling ratio.

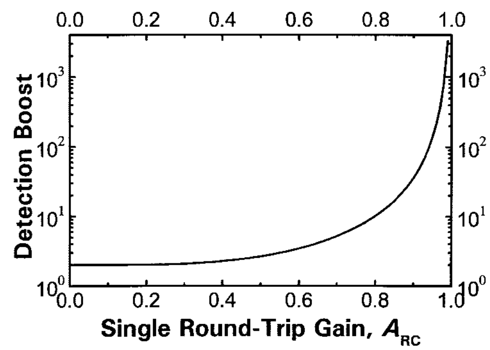


Fig. 6. Detection boost versus single round-trip gain.

by a computer and expressed in the form

$$\frac{x}{1-x}\left\{-n^2x^{2n}+\frac{2nx^{2n}-2(x^2-x^{2(n+1)})/(1-x^2)}{(x^2-1)}+\frac{1-x^{2n}}{1-x^2}\right\}+\frac{y}{1-y}\left\{-n^2y^{2n}+\frac{2ny^{2n}-2(y^2-y^{2(n+1)})/(1-y^2)}{(y^2-1)}+\frac{1-y^{2n}}{1-y^2}\right\}=0 \tag{6}$$

where $x = A_{RC}^2 e^{j\Delta\phi_{ot}}$ and $y = A_{RC}^2 e^{-j\Delta\phi_{\text{opt}}}$.

The optimum phase bias $\Delta\phi_{\text{opt}}$ is plotted against the net single round-trip gain A_{RC} for different numbers of circulation. Fig. 4 shows that the amount of phase bias required for operating the AC-IFOG at the maximum sensitivity point decreases when the net single round-trip gain approaches unity. The optimum biasing point moves faster toward zero when the number of recirculations is large. By substituting ϕ_{opt} into (5), the maximum sensitivity can be obtained. Recalling that $t_o^2 = (1-\gamma)(1-\kappa)e^{-\alpha L}$ from [6], where $\sqrt{(1-\gamma)}$ is the transmission factor of the coupler, $\sqrt{(1-\kappa)}$ is the through coupling ratio of the coupler, α is the fiber propagation loss, and L is the fiber length. If we assume that there is no fiber propagation loss, i.e., $\alpha = 0$, the maximum sensitivity can be plotted for different values of the cross-coupling ratio κ and the single round-trip gain A_{RC}, as shown in Fig. 5. As can be seen, the maximum sensitivity increases tremendously as A_{RC} approaches unity. However, for different coupling ratios κ, the maximum sensitivity profiles are different. Generally, it can be seen that lower coupling ratios have a much higher sensitivity as compared to higher coupling ratios. The drop in the maximum sensitivity at higher coupling ratios may be explained by the fact that less light gets coupled into the sensing coil.

The sensitivity improvement of the proposed AC-IFOG as compared to a conventional IFOG can be indicated by the ratio of the AC-IFOG maximum sensitivity against that of the fundamental IFOG. If we define this as the sensitivity boost factor (SBF), then

$$\text{SBF} = \frac{S_{\max}^{\text{AC-IFOG}}\left(\phi_{\text{opt}}^{\text{AC-IFOG}}\right)}{S_{\max}^{\text{I-FOG}}\left(\phi_{\text{opt}}^{\text{I-FOG}}\right)}. \tag{7}$$

Recalling that the AC-IFOG with $n = 1$ has a response profile equivalent to that of a fundamental IFOG, the maximum sensitivity for this AC-IFOG can thus be found by setting n equal to 1 in (3). Differentiating the equation with respect to $\Delta\phi_R$, we find that its maximum value is

$$S_{\max}^{\text{I-FOG}}\left(\phi_{\text{opt}}^{\text{I-FOG}}\right) = \tfrac{1}{2}\, t_o^2 A_{RC}^2. \tag{8}$$

Using the optimum ϕ_{opt} for both (5) and (8) and substituting them into (7), we obtain (9), shown at the bottom of the page.

In Fig. 6, the SBF is plotted against A_{RC}, and it shows a substantial enhancement as A_{RC} is increased.

The plot shows that, if a sufficiently high single round-trip gain ($A_{RC} > 0.9$) is chosen, a sensitivity boost of at least two orders of magnitude can be attained.

4) Modified CW Mode Operation with Optical Phase Modulation: In the previous section, we discussed the operation of the AC-IFOG in CW mode with the EDFA operating

$$\text{SBF} = \frac{1}{A_{RC}^2}\left\{\frac{(A_{RC}^4-1)}{(1+A_{RC}^4)}\frac{a\,\sin(\Delta\phi_R)}{[1-a\,\text{Cos}(\Delta\phi_R)]^2}+\frac{an}{2}A_{RC}^{2(n+1)}\frac{[a\,\text{Sin}((n-1)\Delta\phi_R)-\text{Sin}(n\Delta\phi_R)]}{[1-a\,\text{Cos}(\Delta\phi_R)]^2}-\frac{a}{2}A_{RC}^{2n}\frac{[a\,\text{Sin}(n\Delta\phi_R)+(an\,\text{Cos}(n\Delta\phi_R)-(n+1))\text{Sin}((n+1)\Delta\phi_R)]}{[1-a\,\text{Cos}(\Delta\phi_R)]^2}\right\} \tag{9}$$

as a switch and the possible enhancement in the maximum sensitivity. In order to make use of the maximum sensitivity, a gyroscope must be biased, and this is generally achieved with optical phase modulation. An optical phase modulator can be placed in one arm near coupler C2, as shown in Fig. 1. Once a modulation signal $\phi(t)$ is produced by the phase modulator, for the primary wave pair, the phase difference between the counterpropagating waves $\Delta\phi_{\rm mod}(t)$ becomes [1]

$$\Delta\phi_{\rm mod}(t) = \phi_{\rm CCW}(t) - \phi_{\rm CW}(t) = \phi(t) - \phi(t-\tau) \quad (10)$$

where $\phi_{\rm CCW}$ and $\phi_{\rm CW}$ represent the phases of the optical waves that propagate in the counterclockwise and clockwise directions in the sensing loop, respectively, and τ (equal to NL/c) is the transit time of light in the fiber loop or the loop transit time. The interference signal becomes

$$I(\Delta\phi_R) = I_o\{1 + \cos[\Delta\phi_R + \Delta\phi_{\rm mod}(t)]\} \quad (11)$$

where $\Delta\phi_R$ is the nonreciprocal phase difference introduced by rotation. For sinusoidal phase modulation with a modulation depth ϕ_o, the biasing phase modulation is $\phi(t) = \phi_o \sin(\omega t)$, and (10) becomes

$$\Delta\phi_{\rm mod}(t) = \phi_o\{\sin(\omega t) - \sin(\omega(t-\tau))\} \quad (12)$$

which can be simplified to

$$\Delta\phi_{\rm mod}(t) = 2\phi_o \sin\left(\omega\frac{\tau}{2}\right)\cos\left(\omega\left(t - \frac{\tau}{2}\right)\right). \quad (13)$$

However, for the mth wave pairs, (13) becomes

$$\Delta\phi_{\rm mod}(t) = 2\phi_o \sin\left(m\omega\frac{\tau}{2}\right)\cos\left(\omega\left(t - m\frac{\tau}{2}\right)\right). \quad (14)$$

Taking into consideration all the wave pairs, (3) can be modified to include the optical phase modulation effect and rewritten as

$$T(\Delta\phi_R) = \frac{1}{2}T_o\sum_{m=1}^{n} A_{RC}^{2m}\Big[1 + \cos\Big(m\Delta\phi_R + 2\varphi_{\rm mod}\cos\Big[\omega\Big(t - m\frac{\tau}{2}\Big)\Big]\Big)\Big] \quad (15)$$

where $\varphi_{\rm mod} = \phi_o \sin(m\omega(\tau/2))$, $T_o = t_o^2$, and n is a number that can approach infinity.

Equation (15) can be expanded into the odd and even components represented by the Bessel function of the first kind

$$T(\Delta\phi_R) = \frac{1}{2}T_o\sum_{m=1}^{n} A_{RC}^{2m}\Bigg\{1 + \Bigg\{J_o(2\varphi_{\rm mod}) + 2\sum_{k=1}^{\infty} J_{2k}(2\varphi_{\rm mod})\cos\Big[2k\omega\Big(t - m\frac{\tau}{2}\Big)\Big]\Bigg\} \cdot\cos(m\Delta\phi_R) - \Bigg\{2\sum_{k=1}^{\infty} J_{2k-1}(2\varphi_{\rm mod}) \cdot\sin\Big[(2k-1)\omega\Big(t - m\frac{\tau}{2}\Big)\Big]\Bigg\}\sin(m\Delta\phi_R)\Bigg\}. \quad (16)$$

We can extract the component at ω by demodulation, i.e., multiplication by $\sin(\omega t)$ followed by lowpass filtering to extract the rotation rate information. In doing so,

$$T(\Delta\phi_R) = -\frac{1}{2}T_o\sum_{m=1}^{n} A_{RC}^{2m} J_1(2\varphi_{\rm mod}) \cdot\cos\Big[m\omega\frac{\tau}{2}\Big]\sin(m\Delta\phi_R). \quad (17)$$

As we are concerned with the maximum sensitivity biasing point, where the differentiation of (17) with respect to $\Delta\phi_R$ is of the largest absolute value, it is differentiated to yield the sensitivity expression

$$\frac{dT(\Delta\phi_R)}{d\Delta\phi_R} = \frac{1}{2}T_o\sum_{m=1}^{n} mA_{RC}^{2m} J_1(2\varphi_{\rm mod}) \cdot\cos\Big[m\omega\frac{\tau}{2}\Big]\cos(m\Delta\phi_R). \quad (18)$$

Assuming that the rotation rate is very small, we can approximate $\cos(m\Delta\phi_R) \approx 1$. The demodulated sensitivity equation can be normalized by $T_o/2$ for the ease of simulation.

a) Effect of A_{RC} on sensitivity: Using the normalized demodulated sensitivity equation, a series of simulations are performed with A_{RC} equal to 0.1, 0.5 and 0.9. In (18), for $m > 10$, it can be shown that the corresponding summation terms become negligibly small, and the simulation is thus limited to just $n = 10$. The objective of the simulation is to examine the dependence of the normalized sensitivity on A_{RC} (the net single round-trip gain), ω (the modulation frequency), and ϕ_o (the modulation depth) so as to determine the maximum sensitivity biasing operating parameters. Fig. 7(a) shows the simulation result when $A_{RC} = 0.1$. As can be seen, the optimum phase modulation bias point varies for each combination of modulation frequency and modulation depth. This figure also shows that, as the modulation depth increases, the required optimum modulation frequency to produce the same level of sensitivity is reduced. Similar trends can be observed in Fig. 7(b) and (c) for $A_{RC} = 0.5$ and $A_{RC} = 0.9$, respectively. From Fig. 7(a)–(c), it can be observed that as A_{RC} increases the maximum sensitivity point is shifted toward the lower modulation frequency region. There is also an overall substantial increase in sensitivity (three orders of magnitude), but the optimum-biasing region becomes narrower. The sharpness of the normalized sensitivity response increases tremendously when the single round-trip gain (A_{RC}) increases. This indicates that the amount of modulation (i.e., the modulation depth) required to bias the gyroscope at its maximum sensitivity has to be reduced to within a narrower range.

b) Effect of n on sensitivity: The variable n indicates the total number of recirculated components that are summed at the detector to give the resultant output. In the previous paragraph, it was pointed out that, for $m > 10$, the corresponding summation terms in (18) are negligibly small. This observation is further examined with the analysis of the effect of n on the normalized sensitivity here. It should be noted that, although the present discussion is made on the assumption that the EDFA is operating in CW mode without switching, the

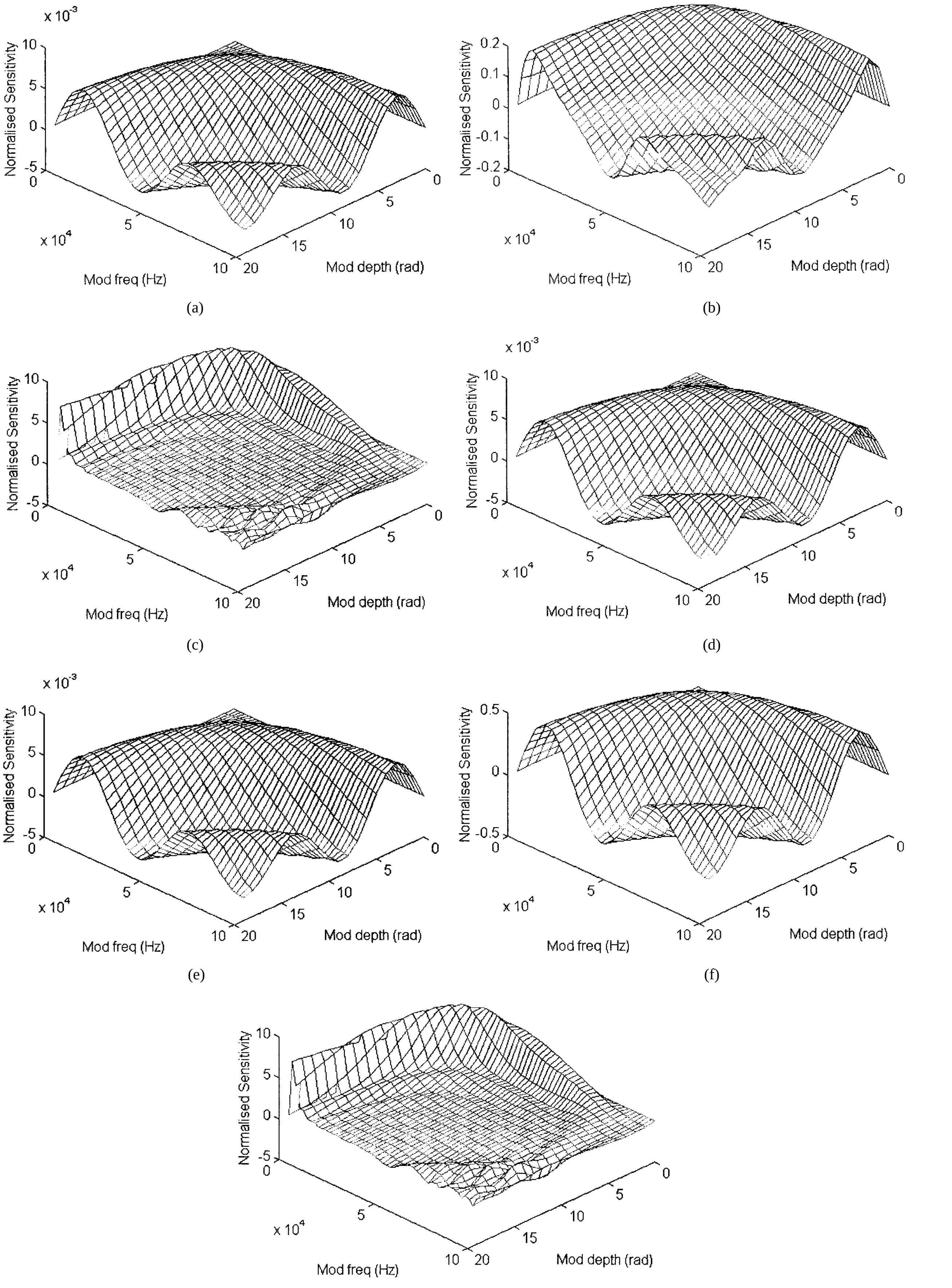


Fig. 7. Normalized sensitivity plot. (a) $A_{RC} = 0.1$, $n = 10$. (b) $A_{RC} = 0.5$, $n = 10$. (c) $A_{RC} = 0.9$, $n = 10$. (d) $A_{RC} = 0.1$, $n = 1$. (e) $A_{RC} = 0.1$, $n = 100$. (f) $A_{RC} = 0.9$, $n = 1$. (g) $A_{RC} = 0.9$, $n = 100$.

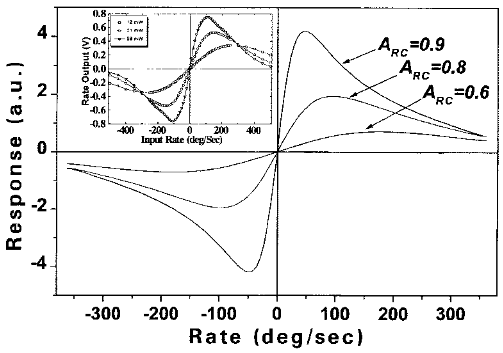


Fig. 8. Open loop response for $A_{RC} = 0.6$, 0.8, 0.9 with all other optical phase modulation parameters optimized.

variable n could in fact be controlled by the switching-on time of the EDFA, if necessary. Fig. 7(d) and (e) show the results of the analysis when $A_{RC} = 0.1$ but $n = 1$ and $n = 100$, respectively. It can be noted that, when the single round-trip gain A_{RC} is small, there is negligible change in the 3-D profile of the normalized sensitivity, even if the total number of recirculation n is varied from 1 to 100. As is expected, when the single round-trip gain is large, say $A_{RC} = 0.9$, a change in n from 1 to 10 causes the normalized sensitivity curve to shift dramatically, as can be seen by comparing Fig. 7(f) and 7(c). However, when a comparison is made between Fig. 7(c) and 7(g), where the n value varies from 10 to 100, it can be found that the change in the profile of the normalized sensitivity is negligible. In fact, this is not surprising because, in (18), the dependence of each summation term on m is dominated by mA_{RC}^{2m}, which would approach zero as m increases beyond 10 simply because A_{RC} must be kept at a value of less than unity to avoid lasing of the gyro coil.

c) Open Loop Response: Using (17), the open loop response is plotted in Fig. 8 by substituting the following parameters in the equation: coil length of 200 m, diameter of 0.16 m, refractive index of 1.46, modulation frequency of 102 kHz. Three curves are plotted for arbitrarily chosen values of A_{RC} of 0.6, 0.8 and 0.9 while the modulation depth $\Delta\phi_{\text{opt}}$ is computed for optimum bias at these points. These are compared with the experimental results obtained in [6]; as shown in the inset, both the experimental results and the simulation have identical profiles at each level of A_{RC} (or EDFA pump power). When A_{RC} or pump power decreases, the peaks begin to shift outwards. A decrease in the response slope at the origin indicates a drop in the detection sensitivity.

II. Theoretical Performance Limits

A. Rotation Detection Limit of AC-IFOG Operating in the Pulse Mode

The theoretical detection limit of the AC-IFOG operating in the pulse mode can be evaluated as follows. With m being the number of recirculations, the pulsed output of the gyro has an envelope that is proportional to $1 + \cos[2\pi f m T_{\text{loop}}]$. The envelope oscillation frequency f is given by $f = D\Omega/N\lambda$, where D is the loop diameter, N the fiber core refractive index, and λ is the optical wavelength in free space. The minimum time delay mT_{loop} required for measuring the frequency f is given by the relation $fmT_{\text{loop}} = 1$. We thus obtain the minimum measurable rotation rate

$$\Omega_{\min} = N\frac{\lambda}{D}\frac{1}{mT_{\text{loop}}}. \tag{19}$$

Considering an ideal device using soliton pulses with in-loop regeneration, the required optical delay $T_{\text{delay-required}}(=mT_{\text{loop}})$ is virtually unlimited as m can approach infinity. Thus, the theoretical AC-IFOG detection limit can be as low as one desires. But this ideal performance cannot be achieved in a practical device because the rotation rate is generally not a constant (varies with time) and the time allowed to collect rotation rate data is always limited to, say, a few milliseconds. For example, consider a device with a loop diameter of $D = 10$ cm and an operating wavelength of $\lambda = 1.5\ \mu$m, and assume that a rotation rate of $\Omega_{\min} \approx 0.001°$/h (comparable to the best IFOG performance) is to be measured. From (19), the required optical delay time (which should be the same as the time required to measure the said rotation rate) is $T_{\text{delay-required}}(=mT_{\text{loop}}) \approx 4500$ s, or 1.25 h. This is far too long a delay time for practical applications. On the other hand, if a loop diameter of $D = 10$ km is used, the required optical delay is reduced to 45 ms. These results show that, practically speaking, high precision operation of such a gyro requires a loop with a really large dimension ($D > 10$ km). Obviously, such a dimension is not suitable for most of the intended applications.

B. Rotation Detection Limit of an AC-IFOG Operating in CW Modes

An AC-IFOG operating in the CW mode has a simpler operation than in the pulse mode because the laser source is not intensity-modulated. Since the rotation detection limit is always an important parameter that forms the grading criteria for all rotation sensors, it has to be investigated theoretically. In this section, we examine and discuss the detection limit with consideration of the on-duration of the EDFA by operating it as though it is an optical switch to selectively allow only a certain number of light recirculations.

It has been shown that the signal and noise propagating in an enclosed loop with an optical amplifier [10] has a limited storage time because of the concurrent growth of ASE noise. The output electrical signal-to-noise (SNR) can be modeled using the equation, as suggested in [8] and originally from [11], as

$$\text{SNR}(m) = \frac{(GL^2\eta I_s)^2}{I_{\text{shot}}^2 + I_{s\text{-sp}}^2 + I_{\text{sp-sp}}^2 + I_{\text{th}}^2} \tag{20}$$

where $I_{\text{shot}}^2 = 2B_e e\eta L(GLI_s + mI_N)$ is the shot noise, $I_{s\text{-}sp}^2 = 4mGL^3\eta^2 I_s I_N B_e/B_o$ and $I_{\text{sp-sp}}^2 = ((mI_N\eta L)^2 B_e(2B_o - B_e))/B_o^2$ are the beat noises, $I_{\text{th}}^2 = 4\kappa T_{\text{eff}} B_e/R_L$ is the thermal noise, $I_s = eP_s/h\nu$ and $I_N = n_{\text{sp}}(G-1)eB_o$ are the received signal and noise photocurrent, respectively, B_o is the optical bandwidth, B_e is the electrical bandwidth, η is the output coupling coefficient, n_{sp} is the spontaneous emission factor, R_L is

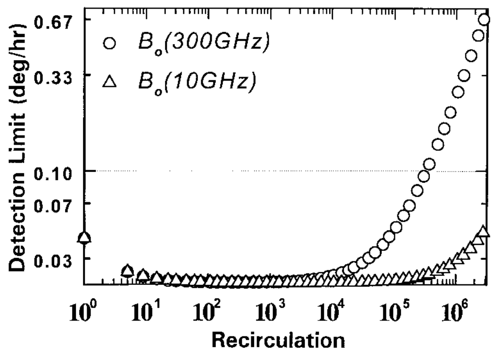


Fig. 9. Detection limits versus the number of circulations.

the load resistance, T_{eff} is the temperature, and P_s is the received optical power.

Equation (20) describes the gradual degradation of the SNR as the number of recirculations (m) increases. When the spontaneous–spontaneous beat noise $I^2_{\text{sp–sp}}$ dominates (increases as a square of m), it also increases the rate of SNR degradation. The important conclusion from this expression is that the optimum number of circulation that gives a unity SNR determines the AC-IFOG detection limit when it is operated with pulse light propagation. However, in the CW light source case, the rotation-induced phase shift is measured by converting it into a change in the intensity of the optical output from the interferometer. In other words, an applied rotation causes an $\Delta\phi_R$, which in turn generates a change in the light intensity that is assumed to be proportional to the rotation at the detector. A problem arises when the intensity of the light falling onto the detector varies due to other reasons such as the unstability of the light source, since this cannot be distinguished from a change in intensity due to a rotation. Therefore, the uncertainty in the measurement of a given rotation rate, i.e., a given $\Delta\phi_R$, might be influenced by the intensity noise of the light source and others. Although there are many ways of compensating for intensity variations of, say, the light source, it is not, however, possible to reduce the effect of photon shot noise because it is a random process. Under ideal conditions, the uncertainty in the measurement of $\Delta\phi_R$ is limited by the total equivalent intensity noise. This uncertainty $\delta(\Delta\phi_R)$ is, therefore, given by [12]

$$\delta(\Delta\phi_R) = \frac{\text{total noise}}{\text{fringe slope}} = \frac{I^2_{\text{shot}} + I^2_{s\text{–}sp} + I^2_{\text{sp–sp}} + I^2_{\text{th}}}{S(\Delta\phi_{\text{opt}})} \tag{21}$$

and is a minimum when the fringe slope is maximum. Since $\Delta\phi_R = (2\pi LD/\lambda c)\Omega$, the uncertainty in the measurement of Ω becomes, after m light circulations,

$$\delta\Omega = \frac{\lambda c}{2\pi m LD}\delta(\Delta\phi_R) \tag{22}$$

where λ is the wavelength of light in free space, L is the length of the coil, D is the diameter of the coil, and m is the number of circulations. Using practical values for our case as reported in our previous paper [6], $\lambda = 1.55$ μm, $L = 200$ m, $D = 0.1$ m, $A_{RC} = 0.9$, cross coupling $\kappa = 0.2$, $B_e = 1$ MHz, $\eta = 1$, $n_{sp} = 1$, $R_L = 50\ \Omega$, $T_{\text{eff}} = 300$K, and $P_s = 1$ mW, the detection limit of the rotation rate Ω can be plotted for $B_o = 300$ GHz and $B_o = 10$ GHz against the number of circulations m, and the result is shown in Fig. 9.

It can be seen that the detection limit improves from an initial value of 0.043°/h to 0.02°/h (set by the signal-spontaneous beat noise) as m approaches 10^2. For $B_o =$ 300 GHz, the spontaneous–spontaneous beat noise component becomes dominant as m exceeds 10^4, and it causes the detection limit to deteriorate. But, for the case of $B_o = 10$ GHz, the low detection limit is extended to beyond 10^5 circulation before it starts to degrade substantially.

III. Conclusion

A novel reentrant fiber-optic gyroscope has been proposed and its principle of operation in the pulse mode as well as in the CW mode has been analyzed theoretically. For pulse mode operation, although the proposed gyroscope offers the advantage of direct frequency readout without the complexity of additional modulation schemes, the analysis shows that a really large coil dimension ($D > 10$ km) is required to achieve low rotation rate detection, and this is impractical for most applications. For CW mode operation, the analysis shows that a scheme making use of optical phase modulation and electronic demodulation is feasible. The sensitivity can be controlled by the gain of the EDFA and can be substantially improved by at least two orders of magnitude as compared to a conventional IFOG. The simulated open-loop response has profiles matching those obtained from the previous experiment. The detection limit is restricted by the beat noises of ASE that build up rapidly in successive circulations. The rate of noise build up depends greatly on the optical bandwidth B_o of the closed-Sagnac loop.

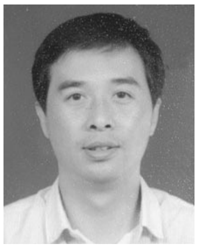
References

[1] V. Vali and R. W. Shorthill, "Fiber ring interferometer," *Appl. Opt.,* vol. 15, pp. 1099–1100, 1976.
[2] H. Lefèvre, *The Fiber-Optic Gyroscope.* Norwell, MA: Artech House, 1993, pp. 19–25.
[3] E. Desurvire, B. Y. Kim, K. Fesler, and H. J. Shaw, "Reentrant fiber raman gyroscope," *J. Lightwave Technol.,* vol. 6, pp. 481–491, Apr. 1988.
[4] C. Y. Liaw, Y. Zhou, and Y. L. Lam, "An erbium fiber based sensitivity booster used by a fiber optic gyroscope," in *1998 Optical Fiber Communication Conf.,* San Jose, CA, Feb. 22–27, pp. 370–371.
[5] J. T. Kringlebotn, "Amplified fiber ring resonator gyro," *IEEE Photon. Technol. Lett.,* vol. 4, pp. 1180, Oct. 1992.
[6] C.-Y. Liaw, Y. Zhou, and Y.-L. Lam, "Characterization of an open-loop interferometric fiber-optic gyroscope with the Sagnac coil closed by an erbium-doped fiber amplifier," *J. Lightwave Technol.,* vol. 16, pp. 2385–2392, Dec. 1998.
[7] G. A. Pavlath and H. J. Shaw, "Reentrant fiber optic rotation sensors," in *Fiber-Optic Rotation Sensors and Related Technologies,* S. Ezekiel and H. J. Arditty, Eds. Berlin, Germany: Springer-Verlag, 1982, p. 364.
[8] D. N. Chen, K. Motoshima, M. M. Downs, and E. Desurvire, "Reentrant Sagnac fiber gyroscope with a recirculating delay line using an erbium-doped fiber amplifier," *IEEE Photon. Technol. Lett.,* vol. 4, pp. 813–815, July 1992.
[9] Yu and A.S. Siddiqui, "Theory of a novel high sensitivity optical fiber gyroscope," *Proc. Inst. Elect. Eng.,* pt. J, vol. 140, no. 2, pp. 150–156, Apr. 1993.
[10] E. Desurvire, M. Tur, and H. J. Shaw, "Signal-to-noise ratio in Raman active fiber systems: Application to recirculating delay lines," *J. Lightwave Technol.,* vol. LT-4, pp. 560–566, May 1986.
[11] N. A. Olsson, "Lightwave systems with optical amplifiers," *J. Lightwave Technol.,* vol. 7, pp. 1071-1083, July 1989.
[12] S. Ezekiel and H. J. Arditty, "Fiber-optic rotation sensors and related technologies," in *Proc. 1st Int. Conf.,* 1981, p. 9.

Chin-Yi Liaw was born in Singapore in 1970. He received the B.Eng. and M.Eng. degrees from Nanyang Technological University (NTU), Singapore, in 1996 and 1999, respectively.

He is now employed by NTU as a Research Associate. His research interests also include image processing and machine pattern recognition.

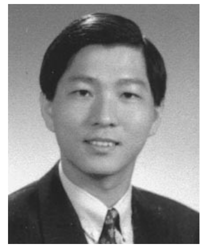
Mr. Liaw was awarded a Gold Medal by the National Science & Technology Board of Singapore for his thesis "Development of a novel interferometric fiber optic gyroscope."



Yan Zhou received the B.Sc. and M.Eng. degrees in China and the Ph.D. degree from Glasgow University, U.K.

In 1992, he joined Nanyang Technological University, Singapore, where he is currently an Associate Professor in the School of Electrical and Electronic Engineering. His research interests include fiber and integrated optics, sol-gel-based photonics materials and devices for optical communication sensing, fiber optic sensors, and optical biosensors.

Dr. Zhou is a member of SPIE.



Yee-Loy Lam (S'91–M'93) received the B.Eng. degree in electrical and electronic engineering and the M.Sc. degree in industrial engineering from the National University of Singapore in 1985 and 1987, respectivel, and the M.S.E. and Ph.D. degrees from the University of Michigan, Ann Arbor, in 1992 and 1993, respectively.

From 1985 to 1990, he was a System Engineer and then a Manager with Rank O'Connor's (S) Pte Ltd., developing and manufacturing building automation products. Thereafter, he joined the School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore, where he is presently Associate Professor, Head of the Microelectronics Division, and Research Coordinator of the Photonics Research Group. His research interests include optoelectronics, sol-gel guided-wave photonics, fiber-optic sensors, lasers, and optical networking.

Dr. Lam is a member of SPIE.